

CPSC 2150 – Spring 26

Ataxx – Project Requirements

F4

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Requirements Analysis

Functional Requirements:

1. As a player, I can move my piece one space adjacent, so that I can get another piece.
2. As a player, I can move my piece 2 spaces adjacent, so I cover more ground trying to capture my opponent .
3. As a player, I can move my piece, so that I can capture my opponents pieces.
4. As a player, I can capture all of my opponents' pieces, so that I can win.
5. As a player, I can see x or o as pieces, so that I can identify which piece is mine.
6. As a player, I can move two spaces away, so that I can perform a jump move.
7. As a player, I can see the amount of pieces each player has, so that I can see who is winning.
8. As a player, I can decide to play again, so that I can try to win.
9. As a player, I can place as many pieces as possible, so that I can try to win.
10. As a player, I can see the columns and rows I can place my piece in, so that I know what play is valid.
11. As a player, I can see the current state of the AtaxxBoard, so that I know where to put my pieces.
12. As a player, I can eliminate my opponents' tokens, so that I can win.
13. As a player, I can see the game identify when it is my turn, so that I know when I move.
14. As a player, I can see that game handled a invalid input carefully, so that the game doesn't crash
15. As a player, I can see the game show whose turn it is, so that I know it's my turn.
16. As a player, I can see the game prompt me to move, so that I can see when it's my turn.
17. As a player, I can move my piece, so that my turn can end.
18. As a player, I can move my piece over another player to a spot, so that I can convert all the pieces around it.
19. As a player, I can see if the move is valid, so that I know if I can move it.
20. As a player, I can input the move, so that I can move the piece.
21. As a player, I can select the piece that I want to move, so that I can move that specific piece.
22. As a player, I can move my piece near my own piece without converting them, so that I can fill up empty space.
23. As a player, I can't select pieces that do not have a valid move, so that I know that piece can't be moves
24. As a player, I can't select other player piece, so that I don't move other player's pieces

Non-Functional Requirements

1. 7X7 board
2. Player 1 starts with 2 pieces .

3. Player 2 starts with 2 pieces.
4. Player 1 has red pieces(in our case "x").
5. Player 2 has blue pieces(in our case "o").
6. Player one has a piece in the NW and in the SE corners.
7. Player two has a piece in the SW and in the NE corners.
8. Board updates whenever player makes valid move
9. Make sure player makes valid move
10. The game ends when one player has no pieces
11. The game ends when board fill up with pieces
12. The game should run on all modern operating systems
13. The game should prompt the player when it's their turn.
14. The game will know which pieces belong to which player
15. The game will turn opponents pieces to players piece if placed adjacent
16. The game should use system resources efficiently
17. The game should load valid saved games correctly
18. The game will make another piece if the piece moved is only moved by one
19. The game requires a system.
20. It needs to be in java